

INFINITE SIMPLE KAZHDAN GROUPS THAT ARE LIMITS OF FINITE SIMPLE GROUPS

SAUERS

ABSTRACT. For every infinite minimal subshift X , the group $G_X = \text{EL}_3(\text{LC}(X, \mathbb{F}_2) \rtimes \mathbb{Z})$ is an infinite, finitely generated, simple group with property (T). It is a limit of finite simple groups $\text{SL}_{3N}(\mathbb{F}_2)$ in the space of marked groups, and their Cayley graphs form a family of expanders. So G_X is locally embeddable into finite groups, sofic and hyperlinear. This answers the question of Brown and Ozawa whether an infinite simple Kazhdan group can be hyperlinear, and Pestov's sofic version of it. A finitely generated group is locally embeddable into finite groups if and only if it is a subgroup of an infinite finitely generated simple Kazhdan group that is a limit of finite simple groups, and one such group contains every recursively presented finitely generated group with this property.

Can an infinite simple group with property (T) be hyperlinear? Brown asked this in 2001 as a question about embeddings into unitary groups of McDuff factors that embed in an ultrapower $\mathcal{R}^\omega$ of the hyperfinite II_1 factor [4, §11, Question 7]. Ozawa stated the hyperlinear form in 2003, noting that the full C^* -algebra of such a group cannot have the local lifting property [36, p. 527], and Pestov's Open question 9.1 adds the sofic form [38]. The groups below answer all three forms positively.

Theorem 1. *Let $X \subseteq A^{\mathbb{Z}}$ be an infinite minimal subshift over a finite alphabet, with shift T , and let $\text{LC}(X, \mathbb{F}_2)$ be the ring of locally constant functions $X \rightarrow \mathbb{F}_2$. Then*

$$G_X = \text{EL}_3(\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$$

is an infinite, finitely generated, simple group with Kazhdan's property (T). It is the limit, in the space of marked groups, of finite simple groups $\text{SL}_{3N}(\mathbb{F}_2)$ whose Cayley graphs, with respect to the images of a fixed generating set, form a family of expanders. So G_X is locally embeddable into finite groups (LEF), sofic and hyperlinear. The same holds for $\text{EL}_n(\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$ for every $n \geq 3$, with $\text{SL}_{nN}(\mathbb{F}_2)$ in place of $\text{SL}_{3N}(\mathbb{F}_2)$.

The ring $\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ is finitely generated, so G_X has property (T), introduced by Kazhdan [24], by the theorem of Ershov and Jaikin-Zapirain [11]. Periodic sequences approximating X give the ring finite models in the matrix algebras $M_N(\mathbb{F}_2)$, as in the proof by Grigorchuk and Medynets that topological full groups of minimal Cantor systems are LEF [16, Theorem 2.6]. So G_X is a limit of the groups $\text{SL}_{3N}(\mathbb{F}_2)$, which form expanders as in Kassabov's construction [23]. The derived subgroup of the topological full group of X is infinite, finitely generated and simple [30], and it is amenable [22]. It is also a subgroup of G_X (Section 3). Simplicity of G_X also follows from Stepanov's theorem on the normal structure of GL_n [43, Theorem 4.4], since the ring is simple [7, Corollary 4.6] and any two of its elements p, q satisfy $pr + qs = 0$ with $(r, s) \neq 0$, by a dimension count on a tower over one small cylinder. Our proof is direct: a nontrivial normal subgroup contains a nontrivial commutator lying in a copy of the finite simple

2020 *Mathematics Subject Classification.* Primary 20E32; Secondary 05C48, 16S35, 20E26, 20F10, 22D55, 37B10, 46L10.

Key words and phrases. Property (T), sofic groups, hyperlinear groups, locally embeddable into finite groups, simple groups, minimal subshifts, expanders.

group $\mathrm{GL}_d(\mathbb{F}_2)$ over a clopen tower, so it contains this group and with it an elementary matrix. For derived topological full groups, Matui showed by a similar argument that a nontrivial normal subgroup meets a simple union of alternating groups on towers [30, Lemma 3.4 and Theorem 4.9].

Hyperlinear groups were named by Rădulescu [40], and sofic groups were introduced by Gromov [19] and named by Weiss [48]. Infinite simple Kazhdan groups arise as quotients of hyperbolic Kazhdan groups [18], [8, Theorem 1.5] and as Kac–Moody lattices, some of which are finitely presented [5]. Thom constructed a finitely generated Kazhdan LEF group that is not residually finite and so lacks the factorization property [45], but his example is not simple. Pestov and Kwiatkowska then wrote that a hope of constructing groups that are not hyperlinear from property (T) remained, as it was unknown whether finitely generated simple Kazhdan groups can be hyperlinear [39].

The proof works for other actions. An action on a Cantor set is topologically free if no nontrivial element fixes a nonempty open set, and $e_{ij}(r) = I_n + rE_{ij}$ is an elementary matrix.

Theorem 2. *Let a finitely generated group Λ act minimally and topologically freely on a Cantor set Z , and let $R = \mathrm{LC}(Z, \mathbb{F}_2) \rtimes \Lambda$ be generated by a finite set $S \ni 1$. Suppose that R is matricial, that is, there are maps $\varphi_k: R \rightarrow M_{N_k}(\mathbb{F}_2)$ with $\varphi_k(1) = I$ such that $\varphi_k(S)$ generates the ring $M_{N_k}(\mathbb{F}_2)$, and for all $r, s \in R$ we have $\varphi_k(r + s) = \varphi_k(r) + \varphi_k(s)$ and $\varphi_k(rs) = \varphi_k(r)\varphi_k(s)$ for large k . Then for every $n \geq 3$, $\mathrm{EL}_n(R)$ is an infinite, finitely generated, simple Kazhdan group. Marked by the $e_{ij}(s)$ with $s \in S$, it is the limit of the finite simple groups $\mathrm{SL}_{nN_k}(\mathbb{F}_2)$ marked by the $e_{ij}(\varphi_k(s))$, whose Cayley graphs form a family of expanders.*

Periodic approximations make $\mathrm{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ matricial, which gives Theorem 1, and lamplighter actions show that the countable subgroups of the groups in Theorem 2 are exactly the countable LEF groups (Corollary 4).

1. PROOF OF THEOREM 2

We write the proof for $n = 3$; for $n \geq 3$ replace 3 by n throughout.

The ring and property (T). Write e_U for the indicator of a clopen set $U \subseteq Z$. The ring $R = \bigoplus_{\xi \in \Lambda} \mathrm{LC}(Z, \mathbb{F}_2) u_\xi$ has the multiplication $(fu_\xi)(f'u_\eta) = f(f' \circ \xi^{-1})u_{\xi\eta}$, so $u_\xi e_U u_\xi^{-1} = e_{\xi U}$. Fix a finite symmetric generating set of Λ , let $|\xi|$ be the word length, and put $B_m = \{\xi \in \Lambda : |\xi| \leq m\}$. Let $G = \mathrm{EL}_3(R)$ be the subgroup of $\mathrm{GL}_3(R)$ generated by the $e_{ij}(r)$ with $i \neq j$ and $r \in R$. With $[g, h] = ghg^{-1}h^{-1}$ and i, j, l distinct,

$$(1) \quad e_{ij}(r + s) = e_{ij}(r)e_{ij}(s), \quad [e_{il}(r), e_{lj}(s)] = e_{ij}(rs).$$

So the $e_{ij}(s)$ with $s \in S$ generate G . By Ershov and Jaikin-Zapirain [11, Theorem 1.1], EL_n of every finitely generated associative unital ring has property (T) for $n \geq 3$. So G has property (T), and it is infinite because $e_{12}(\mathrm{LC}(Z, \mathbb{F}_2))$ is infinite.

Simplicity. Let $1 \neq K \trianglelefteq G$ and $1 \neq g \in K$, and let $w \geq 0$ bound the word lengths of the ξ with u_ξ occurring in the entries of g and g^{-1} . Call a clopen set V small if $V \cap \xi V = \emptyset$ for $\xi \in B_{2w} \setminus \{e\}$ and every $f \circ \xi$, with $\xi \in B_w$ and f a coefficient of an entry of g or g^{-1} , is constant on V . Each $\xi \neq e$ fixes a closed set with empty interior, so the points moved by every $\xi \in B_{2w} \setminus \{e\}$ are dense, and every nonempty clopen set contains a nonempty small one.

Some $h = e_{ij}(e_V)$ with V small does not commute with g . Indeed, g commutes with $e_{ij}(r)$ exactly when $g_{pi}r = 0$ for $p \neq i$, $rg_{jq} = 0$ for $q \neq j$, and $g_{ii}r = rg_{jj}$. For $c = \sum_\xi c_\xi u_\xi$ and small V we have $ce_V = \sum_\xi c_\xi e_{\xi V} u_\xi$ and $e_V c = \sum_\xi e_V c_\xi u_\xi$. If $c_\xi \neq 0$, it equals 1 on a nonempty clopen set U , and a small set $V \subseteq \xi^{-1}U$ gives $ce_V \neq 0$, while a small set $V \subseteq U$ gives $e_V c \neq 0$.

So if g commutes with every such h , then g is diagonal. In $g_{ii}e_V = e_V g_{jj}$ the coefficients at u_ξ with $\xi \neq e$ are supported in the disjoint sets ξV and V , so they vanish, and the same argument shows that the corresponding coefficients of g_{ii} and g_{jj} vanish. The constant coefficients agree on every small V , so they agree everywhere. So $g = cI_3$ with $c \in \text{LC}(Z, \mathbb{F}_2)$, and comparing constant coefficients in $cc^{-1} = 1$ gives $c = 1$ and $g = 1$.

Fix such h and V , and put $d = 3|B_w|$. For $a, b \in B_w$ put $\epsilon_{ab} = e_{aV}u_{ab^{-1}}$. Since $\epsilon_{ab}\epsilon_{a'b'} = e_{a(V \cap b^{-1}a'V)}u_{ab^{-1}a'b'^{-1}}$ and $b^{-1}a' \in B_{2w}$, we get $\epsilon_{ab}\epsilon_{a'b'} = \delta_{ba'}\epsilon_{ab'}$. As $h \neq I_3$, V is nonempty, so the ϵ_{ab} span a ring $A_V \cong M_{|B_w|}(\mathbb{F}_2)$ with unit $e = \sum_a \epsilon_{aa}$, and $y \mapsto (1 - e)I_3 + y$ embeds $\text{GL}_3(A_V) \cong \text{GL}_d(\mathbb{F}_2)$ in $\text{GL}_3(R)$. Its image H_V lies in G , since the transvections $e_{pq}(\epsilon_{ab})$ with $p \neq q$ generate it: the other transvections of $\text{GL}_d(\mathbb{F}_2)$ are commutators of these.

Put $k = [g, h] \in K \setminus \{1\}$. If $a, b \in B_w$ and f, f' are coefficients as above, then

$$f u_a e_V f' u_b = f e_{aV} (f' \circ a^{-1}) u_{ab} \in \{0, \epsilon_{a,b^{-1}}\},$$

since $f \circ a$ and f' are constant on V . The entries of $ghg^{-1} - I_3 = g e_V E_{ij} g^{-1}$ are sums of such products, and $h^{-1} = h = I_3 + \epsilon_{ee}E_{ij}$, so $k - I_3 = (ghg^{-1} - h)h$ and $k^{-1} - I_3 = h(ghg^{-1} - h)$ lie in $M_3(A_V)$. Then $y = eI_3 + (k - I_3)$ and $y' = eI_3 + (k^{-1} - I_3)$ satisfy $yy' = y'y = eI_3$, so $k = (1 - e)I_3 + y \in H_V$. As $K \cap H_V$ is a nontrivial normal subgroup of $H_V \cong \text{GL}_d(\mathbb{F}_2) = \text{PSL}_d(\mathbb{F}_2)$, which is simple as $d \geq 3$, we get $H_V \subseteq K$, and $e_{pq}(e_V) \in K$ for all $p \neq q$.

Finally, $J = \{r \in R : e_{pq}(r) \in K \text{ for all } p \neq q\}$ is a two-sided ideal by (1), since $e_{pq}(sr) = [e_{pl}(s), e_{lq}(r)]$ and $e_{pq}(rs) = [e_{pl}(r), e_{lq}(s)]$ for $l \notin \{p, q\}$. It contains e_V , so it contains every $e_{\xi V} = u_\xi e_V u_\xi^{-1}$. By minimality and compactness finitely many of these sets cover Z , so $1 = 1 - \prod_i (1 - e_{\xi_i V}) \in J$. So $J = R$ and $K = G$.

Finite simple models. A group is LEF [47] if every finite subset embeds injectively into a finite group, preserving all products that stay in that subset. Let F be the free algebra over $\mathbb{F}_2$ on letters τ_s , $s \in S$, and let $\pi: F \rightarrow R$ and $\rho_k: F \rightarrow M_{N_k}(\mathbb{F}_2)$ be the ring homomorphisms with $\pi(\tau_s) = s$ and $\rho_k(\tau_s) = \varphi_k(s)$. Both are onto, so G and $\text{EL}_3(M_{N_k}(\mathbb{F}_2))$ are quotients of $\text{EL}_3(F)$. Here $\text{EL}_3(M_{N_k}(\mathbb{F}_2)) = \text{SL}_{3N_k}(\mathbb{F}_2)$, as transvections generate $\text{SL}_{3N_k}(\mathbb{F}_2)$ and those inside one block are commutators of those between blocks. The group $\text{EL}_3(F)$ is generated by the $e_{ij}(1)$ and $e_{ij}(\tau_s)$, and in both quotients $e_{ij}(1)$ and $e_{ij}(\tau_1)$ have the same image. A word of length λ or less in the $e_{ij}(\tau_s)$ has entries of degree at most λ in F , and $\varphi_k \circ \pi = \rho_k$ on the finitely many such entries for large k . So a word that is trivial in G is trivial in $\text{SL}_{3N_k}(\mathbb{F}_2)$ for large k . Conversely, for a nonprincipal ultrafilter ω on $\mathbb{N}$ the φ_k define a unital ring homomorphism from R to the algebraic ultraproduct $\prod_\omega M_{N_k}(\mathbb{F}_2)$. It maps G to $\prod_\omega \text{SL}_{3N_k}(\mathbb{F}_2)$, and its kernel on G is a normal subgroup that does not contain $e_{12}(1)$, so it is trivial. As this holds for every ω , each nontrivial word is nontrivial in $\text{SL}_{3N_k}(\mathbb{F}_2)$ for all but finitely many k . So these finite simple groups converge to G in the space of marked groups [15, 6], G is LEF, and $N_k \rightarrow \infty$ as G is infinite. Since $\text{EL}_3(F)$ has property (T) [11, Theorem 1.1], its finite quotients form a family of expanders [29, 23]. $\square$

2. PROOF OF THEOREM 1

Let $\mathbb{Z}$ act on X by powers of T , where $(Tx)_t = x_{t+1}$, and write $u = u_1$, so $ufu^{-1} = f \circ T^{-1}$. Minimality and infiniteness imply that T has no periodic points, so the action is topologically free. The ring $R = \text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ is generated by $S = \{1, u, u^{-1}\} \cup \{e_a : a \in A\}$, where $e_a = e_{\{x: x_0=a\}}$, since products of the translates $u^j e_a u^{-j}$ are the cylinder indicators, which span $\text{LC}(X, \mathbb{F}_2)$. We write $\tau_\pm$ and τ_a for the letters of $u^{\pm 1}$ and e_a .

Fix $x \in X$ and $\ell \geq 0$. By minimality the forward orbit of x is dense, so every word of X occurs in $x_{[0, \infty)}$, and $x_{[-\ell, \ell]}$ recurs at arbitrarily large positions. So there is m with $x_{[m-\ell, m+\ell]} = x_{[-\ell, \ell]}$

such that every word of length $2\ell + 1$ of X occurs in $x_{[0,m]}$. The m -periodic sequence y_ℓ that agrees with x on $[0, m)$ then agrees with x on $[-\ell, m + \ell]$. Every window of length $2\ell + 1$ of y_ℓ is a translate of one centered in $[0, m)$, so y_ℓ has the same words of length $2\ell + 1$ as X . Let N_ℓ be its least period, and on $\mathbb{F}_2^{\mathbb{Z}/N_\ell\mathbb{Z}}$ let $P\delta_t = \delta_{t+1}$. For f depending only on coordinates in $[-\ell, \ell]$ let $D_\ell(f)\delta_t = f(T^t y_\ell)\delta_t$, where $f(T^t y_\ell)$ is the value of f at any point of X that agrees with $T^t y_\ell$ on $[-\ell, \ell]$. Put $\varphi_\ell(\sum_j f_j u^j) = \sum_j D_\ell(f_j)P^j$ when every f_j is of this kind, and $\varphi_\ell = 0$ otherwise. Since $PD_\ell(f)P^{-1} = D_\ell(f \circ T^{-1})$ whenever f depends only on coordinates in $[-\ell + 1, \ell - 1]$, for fixed $r, s \in R$ the identities $\varphi_\ell(r + s) = \varphi_\ell(r) + \varphi_\ell(s)$ and $\varphi_\ell(rs) = \varphi_\ell(r)\varphi_\ell(s)$ hold for large ℓ , and $\varphi_\ell(1) = I$. The shifts $T^t y_\ell$, $0 \leq t < N_\ell$, are distinct, so products of the $P^i D_\ell(e_a)P^{-i}$ give every diagonal matrix unit, and with P every matrix unit. So $\varphi_\ell(S)$ generates $M_{N_\ell}(\mathbb{F}_2)$, and Theorem 2 applies. $\square$

Brown's formulation. Let G be a group as in Theorem 2. LEF groups are sofic [38, Example 4.5], and sofic groups are hyperlinear [10, Theorem 2]. So $L(G)$ embeds in $\mathcal{R}^\omega$ [27, 40], [36, Proposition 7.1]. Since G is infinite and simple, its nontrivial conjugacy classes are infinite and G is not residually finite. So $L(G)$ is a II_1 factor, and $L(G) \bar{\otimes} \mathcal{R}$ is a McDuff factor [31] that embeds in $\mathcal{R}^\omega$ and whose unitary group contains G , as in Brown's question. Kirchberg proved that a Kazhdan group with the factorization property is residually finite [27, Theorem 1.1], and that every subgroup of the unitary group of $\mathcal{R}$ has the factorization property [27, Corollary 1.2]. So G does not have the factorization property, it is not a subgroup of the unitary group of $\mathcal{R}$, and $C^*(G)$ does not have the local lifting property [36, p. 527].

3. LEF GROUPS

Every countable group embeds in a finitely generated simple group [14, 42], and even in a simple quotient of a torsion-free hyperbolic Kazhdan group [8, Theorem 1.5], so one simple Kazhdan group contains Higman's universal finitely presented group [21]. Kionke and Schesler proved that every finitely generated residually finite group embeds in a finitely generated simple LEF group [26, Theorem 1.2], so with Wilson's theorem [49] one simple LEF group contains every finitely presented residually finite group. They raised the question which groups embed in finitely generated simple amenable groups. For LEF groups the next corollary answers the analogous question, with a Kazhdan host. Its second statement is the analogue for LEF groups of the theorem of Boone and Higman that a finitely generated group has solvable word problem if and only if it embeds in a simple subgroup of a finitely presented group [3], in the form of Thompson's strengthening, where the simple group is finitely generated and has solvable word problem [46].

Corollary 3. *A finitely generated group is LEF if and only if it is a subgroup of an infinite, finitely generated, simple group with property (T) that is a limit of finite simple groups $\text{SL}_N(\mathbb{F}_2)$ in the space of marked groups, whose Cayley graphs form a family of expanders. It is LEF with solvable word problem if and only if it is a subgroup of such a group with solvable word problem.*

Corollary 4. *Every countable LEF group is a subgroup of such a group. A set of finitely generated groups lies in one such group if and only if its members are LEF and represent countably many isomorphism classes. So one such group contains every recursively presented finitely generated LEF group, for instance every finitely presented residually finite group and every G_Y whose language is recursive.*

Proof. Limits of finite groups in the space of marked groups are LEF, and so are their subgroups. Conversely, let $\Gamma = \{\gamma_1, \gamma_2, \dots\}$ be a countable LEF group, replaced by $\Gamma \times \mathbb{Z}$ so that it is infinite. There are finite groups Q_k and maps $\mu_k: \Gamma \rightarrow Q_k$ that are injective on $\{1, \gamma_1, \dots, \gamma_k\}$

and preserve the products that stay in this set. On $Q_k \times \mathbb{Z}/m_k$, with $m_k = |Q_k|$, let t_k add 1 to the second coordinate, and let f_{jk} multiply the first coordinate of (x, i) from the left by $\mu_k(\gamma_j)^i$, which is well defined as $|Q_k|$ divides m_k . Then $[f_{jk}, t_k]$ multiplies the first coordinate by $\mu_k(\gamma_j)$. On $Q_k \times \mathbb{Z}/m_k \times \mathbb{Z}/4^{k+1}$ let b_k add 1 to the last coordinate, and let a_k act on the level i by f_{jk} if $i = 2^{2j-1}$ and by t_k if $i = 4^j$, for $j \leq k$, and trivially on the other levels. Then $b_k^{-n} a_k b_k^n$ acts on the level i as a_k acts on the level $i + n$. As distinct pairs of powers of 2 have distinct differences, all below a quarter of the modulus, the word $w_j = [b^{-2^{2j-1}} a b^{2^{2j-1}}, b^{-4^j} a b^{4^j}]$ evaluated at (a_k, b_k) acts by $[f_{jk}, t_k]$ on the level 0 and trivially on the other levels, for $j \leq k$. In the algebraic ultraproduct over ω of the symmetric groups of these sets, sending γ to the class of the permutations multiplying the first coordinate on the level 0 by $\mu_k(\gamma)$ and acting trivially elsewhere is a homomorphism, injective as $\mu_k(\gamma) \neq 1$ for $\gamma \neq 1$ and large k . It sends γ_j to $w_j(a, b)$, where $a = (a_k)_k$ and $b = (b_k)_k$, so $\Gamma \leq [\Delta, \Delta]$ for the group Δ generated by a and b . Finitely many products and inequalities in the ultraproduct hold in some coordinate, so Δ is infinite, finitely generated and LEF.

If Γ is generated by a finite set E , another choice of Δ keeps a solvable word problem. On $\Gamma \times \mathbb{Z}$, whose second coordinate is the level, let β raise the level by 1, and let h_γ multiply the first coordinate by γ on the levels $m \geq 0$. Then $[\beta, h_\gamma]$ multiplies the level 0 by γ^{-1} , so $\Delta = \langle \beta, h_\gamma : \gamma \in E \rangle$ is infinite and $\Gamma \leq [\Delta, \Delta]$. An element of Δ of length at most r is $f\beta^\ell$ with $|\ell| \leq r$, where f multiplies the level m by an element $f(m)$ of length at most r , with $f(m) = 1$ for $m < -r$ and $f(m) = f(r)$ for $m > r$. Let ψ map the ball of radius $2r$ of Γ injectively into a finite group Q , preserving the products that stay in it, and let θ , with $\theta(v)_{-3r} = 1$ and $\theta(v)_m = v_{m-1}$ otherwise, be the isomorphism from $\{v \in Q^{[-3r, 3r]} : v_{3r-1} = v_{3r}\}$ onto $\{v \in Q^{[-3r, 3r]} : v_{-3r} = 1\}$. Transversals give a bijection κ of $Q^{[-3r, 3r]}$ with $\kappa(vx) = \theta(v)\kappa(x)$ for v in the domain of θ . Let each $v \in Q^{[-3r, 3r]}$ act on $Q^{[-3r, 3r]} \times \mathbb{Z}/5r$ by left multiplication on the first coordinate, and put $\zeta(x, i) = (\kappa(x), i + 1)$, so that $\zeta v \zeta^{-1} = \theta(v)$ on the domain of θ . As f is trivial below $-r$ and constant from r on, $\zeta^\ell(\psi(f(m)))_m \zeta^{-\ell} = (\psi(f(m - \ell)))_m$ for $|\ell| \leq r$. So $f\beta^\ell \mapsto (\psi(f(m)))_{|m| \leq 3r} \zeta^\ell$ preserves the products that stay in the ball of radius r , and it is injective there, as $5r > 2r$ and f is determined by its values for $|m| \leq 3r$. So Δ is LEF, and the rest of the proof applies to it.

Let Δ act on $\Omega = \mathbb{F}_2^\Delta$ by $(\delta x)(h) = x(\delta^{-1}h)$, and let Λ be the group of maps $x \mapsto \delta x + f$ with $\delta \in \Delta$ and f finitely supported. It is generated by Δ and the map z adding 1 at e , and its orbits contain the dense sets $x + \bigoplus_\Delta \mathbb{F}_2$. If $(\delta, f) \neq (e, 0)$, the fixed points of $x \mapsto \delta x + f$ satisfy $x(h) = x(\delta^{-1}h) + f(h)$ for all h . In every cylinder over a finite set W this fails at some point, for $h \in \text{supp } f$ if $\delta = e$ and for $h \notin W \cup \delta W \cup \text{supp } f$ if $\delta \neq e$. So the action is minimal and topologically free. The ring $R_\Delta = \text{LC}(\Omega, \mathbb{F}_2) \rtimes \Lambda$ is generated by 1, the units $u_s^{\pm 1}$ for the generators s of Λ , and the indicator e_U of $U = \{x : x(e) = 1\}$.

Let $\sigma_n : \Delta \rightarrow Q'_n$ be finite models of Δ that are injective on the ball of radius n and preserve the products that stay in it, with $\sigma_n(e) = e$ and Q'_n generated by the images of the generators. Let $x \mapsto \delta x + f$ act on $Y_n = \mathbb{F}_2^{Q'_n}$ by $y \mapsto \sigma_n(\delta)y + \sum_{h \in \text{supp } f} 1_{\sigma_n(h)}$, where $(qy)(q') = y(q^{-1}q')$, and attach to y the point $y \circ \sigma_n$ of Ω . For fixed elements of Λ and large n , these permutations multiply as in Λ and, uniformly in y , move the attached points near e as Λ moves the points of Ω . Let $D_n(f)$ be the diagonal matrix on $\mathbb{F}_2^{Y_n}$ with entry $f(y \circ \sigma_n)$ at y , let P_ξ be the permutation matrix of $\xi \in \Lambda$, and put $\varphi_n(\sum_\xi f_\xi u_\xi) = \sum_\xi D_n(f_\xi) P_\xi$. For fixed f and ξ and large n ,

$$P_\xi D_n(f) P_\xi^{-1} = D_n(f \circ \xi^{-1}),$$

so for fixed $r, s \in R_\Delta$ and large n , φ_n is additive and multiplicative on r, s , and $\varphi_n(1) = I$. The conjugates of $D_n(e_U)$ by products of the P_s are the coordinate functions of Y_n , and the

conjugates of P_z are the translations by the 1_q , so the images of the generators generate $M_{2|Q'_n|}(\mathbb{F}_2)$. By Theorem 2, $G_\Delta = \text{EL}_3(R_\Delta)$ is an infinite finitely generated simple Kazhdan group, and the limit of finite simple groups $\text{SL}_{3N}(\mathbb{F}_2)$ whose Cayley graphs form expanders.

Finally, $\xi \mapsto \text{diag}(u_\xi, 1, 1)$ is an injective homomorphism $\Lambda \rightarrow \text{GL}_3(R_\Delta)$. It maps $[\Lambda, \Lambda]$ into G_Δ , since over $\mathbb{F}_2$ every unit v satisfies $\text{diag}(v, v^{-1}, 1) = e_{12}(v)e_{21}(v^{-1})e_{12}(v)e_{12}(1)e_{21}(1)e_{12}(1)$, and the product of these matrices for $v = c, c'$ and $(c'c)^{-1}$ is $\text{diag}(cc'c^{-1}c'^{-1}, 1, 1)$. So G_Δ contains $\Gamma \leq [\Delta, \Delta] \leq [\Lambda, \Lambda]$. This proves the first statements of Corollaries 3 and 4. For the second statement of Corollary 4, apply the first to the restricted direct sum of representatives of countably many isomorphism classes of finitely generated LEF groups, which is a countable LEF group, as finite subsets lie in finite products. Conversely, a countable group has countably many finitely generated subgroups. There are countably many recursive presentations.

Now let Γ be finitely generated, LEF and with solvable word problem, and let Δ be the second choice. A word in the $\beta^{\pm 1}$ and $h_\gamma^{\pm 1}$ gives ℓ , the levels where f changes, which are the exponent sums of β before the letters $h_\gamma^{\pm 1}$, and the values of f there as words in E . It is trivial if and only if $\ell = 0$ and these values are trivial, so Δ has solvable word problem. An element $x \mapsto \delta x + c$ of Λ is given by a word for δ and words for the lamps of c , so the word problem of Δ decides equality in Λ and computes the values of c . A word in the generators of G_Δ multiplies out to a matrix with entries $\sum_\xi p_\xi u_\xi$, where each p_ξ is a polynomial over $\mathbb{F}_2$ in finitely many coordinates $x(\delta)$, as $u_\xi e_U u_\xi^{-1}$ is the indicator of $\{x : x(\delta) = 1 + c(\delta)\}$ for $\xi : x \mapsto \delta x + c$. After collecting equal ξ and equal coordinates, the word is trivial if and only if every coefficient of its difference from I_3 vanishes at every assignment of its finitely many variables, as Ω is the full shift. So G_Δ has solvable word problem. Conversely, finitely generated subgroups of groups with solvable word problem have solvable word problem, and LEF passes to subgroups. This proves the second statement of Corollary 3. $\square$

Applied to the finitely generated residually finite non-exact groups of Osajda [35], which are LEF, Corollary 3 gives infinite simple Kazhdan LEF groups that are not exact, since exactness, that is, property A, passes to subgroups.

For g in the topological full group of X put $w_g = \sum_n u^n e_{\{x: gx=T^n x\}}$. Then $g \mapsto w_g$ is an injective homomorphism into $\text{GL}_1(\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$, so $g \mapsto \text{diag}(w_g, 1, 1)$ is an injective homomorphism, and by the same identities it embeds the derived subgroup, which is generated by commutators, in G_X .

Every group as in the last statement of Corollary 4 has unsolvable word problem: a recursive bound on the running time of an algorithm for it would bound, up to a linear change of word length, the time complexity of the word problems of all finitely presented residually finite groups, and these can be as hard as the membership problem of any recursive set [25, Theorem 4.21]; compare [41, Proposition 26 and the proof of Lemma 27].

The groups G_X do not suffice here: $\text{SL}_3(\mathbb{Z})$ lies in no G_X . The ring $M_m(\text{LC}(X, \mathbb{F}_2) \rtimes \mathbb{Z})$ acts faithfully on $\bigoplus_{x \in X, t \in \mathbb{Z}} \mathbb{F}_2^m$ by $f\delta_{x,t} = f(T^t x)\delta_{x,t}$ and $u\delta_{x,t} = \delta_{x,t+1}$. If the generators of a subgroup of GL_m involve only u^j with $|j| \leq w$, where $w \geq 1$, and $|g^n|/n \rightarrow 0$, then for v supported at one (x, t) the $g^n v$, $n < N$, span fewer than N dimensions for large N , uniformly in v . So each such v is killed by a polynomial over $\mathbb{F}_2$ of degree below N ; there are finitely many of these, their product kills g , and g has finite order. In the Heisenberg subgroup of $\text{SL}_3(\mathbb{Z})$, $[a, b]$ has infinite order and $[a, b]^{n^2} = [a^n, b^n]$. Cyr, Franks, Kra and Petite use distortion similarly for automorphism groups of shifts [9].

4. WORD PROBLEMS AND FACTORS

Corollary 5. *The word problem of G_X has the Turing degree of the language $L(X)$ of X . In particular G_X has solvable word problem if and only if $L(X)$ is recursive, as for the Fibonacci subshift $X_{(3-\sqrt{5})/2}$ below. Every Turing degree occurs, and there are continuum many groups G_X , none of which is isomorphic to a subgroup of another.*

Proof. Multiplying out a word in the generators in $\text{LC}(A^{\mathbb{Z}}, \mathbb{F}_2) \rtimes \mathbb{Z}$, which maps onto R , gives a matrix with entries $\sum_j f_j u^j$, each f_j given by a table on the words of some length, using $uf = (f \circ T^{-1})u$. The word is trivial in G_X if and only if the tables of its difference from I_3 vanish on $L(X)$, so $L(X)$ computes the word problem. Conversely, by (1) a word for $e_{12}(\prod_{t < m} u^{-t} e_{v_t} u^t)$ is computable from $v = v_0 \cdots v_{m-1}$, and it is trivial if and only if $v \notin L(X)$.

For irrational $\alpha \in (0, 1)$ let X_α be the infinite minimal Sturmian subshift, the closure of the codings $c(\theta)$, $\theta \in [0, 1)$, with $c(\theta)_t = \lfloor \theta + (t+1)\alpha \rfloor - \lfloor \theta + t\alpha \rfloor$, which is 1 exactly for θ in the arc $[-(t+1)\alpha, -t\alpha]$ modulo 1 [32, 20], [28, Chapter 2]. Passing to the closure adds no words, so the words of length m of X_α are the constant values of $c(\theta)_{[0, m]}$ on the arcs between the distinct points $-j\alpha \bmod 1$, $0 \leq j \leq m$, and α computes $L(X_\alpha)$. Such a word has $\lfloor \theta + m\alpha \rfloor$ ones, within 1 of $m\alpha$, so $L(X_\alpha)$ computes α . Every $S \subseteq \mathbb{N}$ has the degree of $[0; 1 + \chi_S(0), 1 + \chi_S(1), \dots]$, since each digit is decided by one strict comparison with a rational computed from earlier digits. If a finitely generated group H embeds in a finitely generated group G , a word for H becomes a word for G by substituting for its generators, so the word problem of H reduces to that of G . The Turing degrees contain an antichain of size continuum [34, Chapter V], and the groups G_{X_α} with α of these degrees are as claimed. $\square$

For derived topological full groups, Grigorchuk and Medynets proved that the word problem is decidable if and only if $L(X)$ is recursive [17, Theorem 1.1(3)].

For a fixed alphabet, every G_X is a simple quotient of the Kazhdan group $\text{EL}_3(F)$, and $e_{12}(\prod_{t < m} \tau_-^t \tau_{v_t} \tau_+^t)$ lies in the kernel if and only if $v \notin L(X)$, so different subshifts give different kernels. The proof of [37, Theorem 2] then shows that the unitary group of a separable II_1 factor contains G_X for only countably many X . So the II_1 factors $L(G_X)$ with $X \subseteq \{0, 1\}^{\mathbb{Z}}$, which embed in $\mathcal{R}^\omega$, fall into continuum many isomorphism classes, each containing $L(G_X)$ for only countably many X . Nicoara, Popa and Sasyk proved the same kind of statement for twisted group factors of $\mathbb{Z}^2 \rtimes \Gamma$ with $\Gamma \leq \text{SL}_2(\mathbb{Z})$ nonamenable, which also embed in $\mathcal{R}^\omega$ [33, Theorem 0.1].

5. QUESTIONS

A finitely presented LEF group is residually finite [44, 47], and the infinite simple group G_X is not, so G_X is not finitely presented. Finitely presented infinite simple groups with property (T) exist [5]. Is there one that is sofic, or at least hyperlinear? A sofic example would not be LEF, and it would also answer Open problem 6.1 of Alekseev and Thom [1], which asks for finitely presented sofic groups with property (T) that are not residually finite. Is every sofic infinite simple Kazhdan group LEF?

Is every finitely generated sofic group a subgroup of an infinite finitely generated simple Kazhdan sofic group, as in Corollary 3?

If (X, T) is topologically conjugate to (Y, S) or to (Y, S^{-1}) , then $G_X \cong G_Y$. For topological full groups the converse holds [13, 2]. Does $G_X \cong G_Y$ imply that X and Y are flip conjugate, or at least strongly orbit equivalent [12]? Does $L(G_X) \cong L(G_Y)$ imply $G_X \cong G_Y$?

ORIGIN AND AUTHORSHIP

Claude (Anthropic) found the construction and original proofs under the author's direction on September 12, 2026, and wrote the manuscript. The author is responsible for the final manuscript. Proof records are in the [project repository](#).

REFERENCES

- [1] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, [arXiv:2608.05362](#) (2026).
- [2] S. Bezuglyi and K. Medynets, *Full groups, flip conjugacy, and orbit equivalence of Cantor minimal systems*, *Colloq. Math.* **110** (2008), 409–429. [doi:10.4064/cm110-2-6](#).
- [3] W. W. Boone and G. Higman, *An algebraic characterization of groups with soluble word problem*, *J. Austral. Math. Soc.* **18** (1974), 41–53. [doi:10.1017/S1446788700019108](#).
- [4] N. P. Brown, *Tracial invariants, classification and II_1 factor representations of Popa algebras*, [arXiv:math/0111286](#) (2001).
- [5] P.-E. Caprace and B. Rémy, *Simplicity and superrigidity of twin building lattices*, *Invent. Math.* **176** (2009), 169–221. [doi:10.1007/s00222-008-0162-6](#).
- [6] C. Champetier, *L'espace des groupes de type fini*, *Topology* **39** (2000), 657–680. [doi:10.1016/S0040-9383\(98\)00063-9](#).
- [7] L. O. Clark and C. Edie-Michell, *Uniqueness theorems for Steinberg algebras*, *Algebr. Represent. Theory* **18** (2015), 907–916. [doi:10.1007/s10468-015-9522-2](#).
- [8] R. Coulon and F. Fournier-Facio, *Infinite simple characteristic quotients*, preprint (2023). [arXiv:2312.11684](#).
- [9] V. Cyr, J. Franks, B. Kra, and S. Petite, *Distortion and the automorphism group of a shift*, *J. Mod. Dyn.* **13** (2018), 147–161. [doi:10.3934/jmd.2018015](#).
- [10] G. Elek and E. Szabó, *Hyperlinearity, essentially free actions and L^2 -invariants. The sofic property*, *Math. Ann.* **332** (2005), 421–441. [doi:10.1007/s00208-005-0640-8](#).
- [11] M. Ershov and A. Jaikin-Zapirain, *Property (T) for noncommutative universal lattices*, *Invent. Math.* **179** (2010), 303–347. [doi:10.1007/s00222-009-0218-2](#).
- [12] T. Giordano, I. F. Putnam, and C. F. Skau, *Topological orbit equivalence and C^* -crossed products*, *J. Reine Angew. Math.* **469** (1995), 51–111. [doi:10.1515/crll.1995.469.51](#).
- [13] T. Giordano, I. F. Putnam, and C. F. Skau, *Full groups of Cantor minimal systems*, *Israel J. Math.* **111** (1999), 285–320. [doi:10.1007/BF02810689](#).
- [14] A. P. Gorjuškin, *Imbedding of countable groups in 2-generated simple groups*, *Math. Notes* **16** (1974), 725–727. [doi:10.1007/BF01105577](#).
- [15] R. I. Grigorchuk, *Degrees of growth of finitely generated groups, and the theory of invariant means*, *Math. USSR-Izv.* **25** (1985), 259–300. [doi:10.1070/IM1985v025n02ABEH001281](#).
- [16] R. Grigorchuk and K. Medynets, *On algebraic properties of topological full groups*, *Sb. Math.* **205** (2014), 843–861. [doi:10.1070/SM2014v205n06ABEH004400](#).
- [17] R. Grigorchuk and K. Medynets, *Presentations of topological full groups by generators and relations*, *J. Algebra* **500** (2018), 46–68. [doi:10.1016/j.jalgebra.2016.10.027](#).
- [18] M. Gromov, *Hyperbolic groups*, in: *Essays in group theory*, *Math. Sci. Res. Inst. Publ.* **8**, Springer, New York (1987), 75–263. [doi:10.1007/978-1-4613-9586-7_3](#).
- [19] M. Gromov, *Endomorphisms of symbolic algebraic varieties*, *J. Eur. Math. Soc.* **1** (1999), 109–197. [doi:10.1007/PL00011162](#).
- [20] G. A. Hedlund, *Sturmian minimal sets*, *Amer. J. Math.* **66** (1944), 605–620. [doi:10.2307/2371769](#).
- [21] G. Higman, *Subgroups of finitely presented groups*, *Proc. Roy. Soc. London Ser. A* **262** (1961), 455–475. [doi:10.1098/rspa.1961.0132](#).
- [22] K. Juschenko and N. Monod, *Cantor systems, piecewise translations and simple amenable groups*, *Ann. of Math. (2)* **178** (2013), 775–787. [doi:10.4007/annals.2013.178.2.7](#).
- [23] M. Kassabov, *Universal lattices and unbounded rank expanders*, *Invent. Math.* **170** (2007), 297–326. [doi:10.1007/s00222-007-0064-z](#).
- [24] D. A. Kazhdan, *Connection of the dual space of a group with the structure of its closed subgroups*, *Funct. Anal. Appl.* **1** (1967), 63–65. [doi:10.1007/BF01075866](#).
- [25] O. Kharlampovich, A. Myasnikov, and M. Sapir, *Algorithmically complex residually finite groups*, *Bull. Math. Sci.* **7** (2017), 309–352. [doi:10.1007/s13373-017-0103-z](#).

- [26] S. Kionke and E. Schesler, *From telescopes to frames and simple groups*, J. Comb. Algebra, online first (2024). doi:10.4171/JCA/103. arXiv:2304.09307.
- [27] E. Kirchberg, *Discrete groups with Kazhdan's property T and factorization property are residually finite*, Math. Ann. **299** (1994), 551–563. doi:10.1007/BF01459798.
- [28] M. Lothaire, *Algebraic combinatorics on words*, Encyclopedia Math. Appl. **90**, Cambridge Univ. Press, Cambridge, 2002. doi:10.1017/CB09781107326019.
- [29] G. A. Margulis, *Explicit construction of a concentrator* (Russian), Problemy Peredachi Informatsii **9** (1973), no. 4, 71–80.
- [30] H. Matui, *Some remarks on topological full groups of Cantor minimal systems*, Internat. J. Math. **17** (2006), 231–251. doi:10.1142/S0129167X06003448.
- [31] D. McDuff, *Central sequences and the hyperfinite factor*, Proc. London Math. Soc. (3) **21** (1970), 443–461. doi:10.1112/plms/s3-21.3.443.
- [32] M. Morse and G. A. Hedlund, *Symbolic dynamics II. Sturmian trajectories*, Amer. J. Math. **62** (1940), 1–42. doi:10.2307/2371431.
- [33] R. Nicoara, S. Popa and R. Sasyk, *On II_1 factors arising from 2-cocycles of w -rigid groups*, J. Funct. Anal. **242** (2007), 230–246. doi:10.1016/j.jfa.2006.05.015.
- [34] P. Odifreddi, *Classical recursion theory*, Vol. I, Stud. Logic Found. Math. **125**, North-Holland, Amsterdam, 1989.
- [35] D. Osajda, *Residually finite non-exact groups*, Geom. Funct. Anal. **28** (2018), 509–517. doi:10.1007/s00039-018-0432-4.
- [36] N. Ozawa, *About the QWEP conjecture*, Internat. J. Math. **15** (2004), 501–530; preprint arXiv:math/0306067 (2003). doi:10.1142/S0129167X04002417.
- [37] N. Ozawa, *There is no separable universal II_1 -factor*, Proc. Amer. Math. Soc. **132** (2004), 487–490. doi:10.1090/S0002-9939-03-07127-2.
- [38] V. G. Pestov, *Hyperlinear and sofic groups: a brief guide*, Bull. Symbolic Logic **14** (2008), 449–480. doi:10.2178/bsl/1231081461.
- [39] V. G. Pestov and A. Kwiatkowska, *An introduction to hyperlinear and sofic groups*, in: Appalachian Set Theory 2006–2012, London Math. Soc. Lecture Note Ser. **406**, Cambridge Univ. Press, Cambridge (2012), 145–186. doi:10.1017/CB09781139208574.006.
- [40] F. Rădulescu, *The von Neumann algebra of the non-residually finite Baumslag group $\langle a, b \mid ab^3a^{-1} = b^2 \rangle$ embeds into R^ω* , Theta Ser. Adv. Math. **9**, Theta, Bucharest (2008), 173–185; preprint arXiv:math/0004172 (2000).
- [41] E. Rauzy, *Remarks and problems about algorithmic descriptions of groups*, Internat. J. Algebra Comput. **35** (2025), 1019–1053. doi:10.1142/S0218196725500328.
- [42] P. E. Schupp, *Embeddings into simple groups*, J. London Math. Soc. (2) **13** (1976), 90–94. doi:10.1112/jlms/s2-13.1.90.
- [43] A. V. Stepanov, *On the normal structure of the general linear group over a ring*, Zap. Nauchn. Sem. S.-Peterburg. Otdel. Mat. Inst. Steklov. (POMI) **236** (1997), 166–182; English transl., J. Math. Sci. **95** (1999), 2146–2155. doi:10.1007/BF02169976.
- [44] A. M. Stepin, *A remark on the approximability of groups*, Vestnik Moskov. Univ. Ser. I Mat. Mekh. (1984), no. 4, 85–87; English transl., Moscow Univ. Math. Bull. **39** (1984), no. 4, 90–93.
- [45] A. Thom, *Examples of hyperlinear groups without factorization property*, Groups Geom. Dyn. **4** (2010), 195–208. doi:10.4171/GGD/80.
- [46] R. J. Thompson, *Embeddings into finitely generated simple groups which preserve the word problem*, in: Word problems II, Stud. Logic Found. Math. **95** (1980), 401–441. doi:10.1016/S0049-237X(08)71348-X.
- [47] A. M. Vershik and E. I. Gordon, *Groups that are locally embeddable in the class of finite groups*, Algebra i Analiz **9** (1997), no. 1, 71–97; English transl., St. Petersburg Math. J. **9** (1998), no. 1, 49–67.
- [48] B. Weiss, *Sofic groups and dynamical systems*, Sankhyā Ser. A **62** (2000), 350–359.
- [49] J. S. Wilson, *Embedding theorems for residually finite groups*, Math. Z. **174** (1980), 149–157. doi:10.1007/BF01293535.